Year 11 Set 2 · Higher · grade 5–6 · Lesson 7

Direct & inverse proportion

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Today I will learn to...

Goals

1. Solve direct proportion problems

2. Solve inverse proportion problems

Do now: retrieval practice

4 mins

1 P(none) = 0.36. P(at least one)?

0.64

2 $3^2 \times 4$

36

3 $\sqrt{(36 \div 4)}$

3

4 Rearrange $y = kx$ for k

$k = y/x$

Objective 1

Solve direct proportion problems

Key idea

Learn

!

$y \propto x$ means $y = kx$. Use the given values to find k, write the formula, then use it.

Objective 1: Solve direct proportion problems

I do: example 1

I do

$y \propto x$. $y = 12$ when $x = 4$. Find y when $x = 7$.

1. Write $y = kx$
2. Substitute: $12 = k \times 4$, so $k = 3$
3. Formula: $y = 3x$
4. When $x = 7$: $y = 3 \times 7 = 21$

I do: example 2

I do

$y \propto x^2$. $y = 50$ when $x = 5$. Find y when $x = 3$.

1. Write $y = kx^2$
2. Substitute: $50 = k \times 25$, so $k = 2$
3. Formula: $y = 2x^2$
4. When $x = 3$: $y = 2 \times 9 = 18$

I do: example 3

I do

$d \propto \sqrt{h}$. $d = 12$ when $h = 16$. Find d when $h = 25$.

1. Write $d = k\sqrt{h}$
2. Substitute: $12 = k \times \sqrt{16} = 4k$, so $k = 3$
3. Formula: $d = 3\sqrt{h}$
4. When $h = 25$: $d = 3 \times 5 = 15$

Mini whiteboards

We do

1. $y \propto x$, $y = 20$ when $x = 5$. y when $x = 8$?

Answer: 32

2. $y \propto x^3$, $y = 16$ when $x = 2$. y when $x = 3$?

Answer: 54

3. $y \propto x^2$, $y = 12$ when $x = 2$. y when $x = 5$?

Answer: 75

4. $y \propto x$, $y = 7$ when $x = 2$. x when $y = 21$?

Answer: 6

You do: choose your challenge

You do

Bronze

$y \propto x$. $y = 15$ when $x = 3$. Find y when $x = 10$.

Silver

$y \propto x^2$. $y = 36$ when $x = 3$. Find y when $x = 5$.

Gold

Cost is proportional to weight. 2.5 kg costs £6. What does 4 kg cost?

Platinum

$y \propto x^3$. $y = 54$ when $x = 3$. Find x when $y = 250$.

You do: answers

Check

Bronze	50
Silver	100
Gold	£9.60
Platinum	$k = 2$, $x^3 = 125$, $x = 5$

If you finish early

Extra

Bronze

$y \propto x$. $y = 18$ when $x = 6$. Find y when $x = 11$.

33

Silver

$y \propto x^2$. $y = 20$ when $x = 2$. Find y when $x = 3$.

45

Gold

$d \propto t^2$. $d = 45$ when $t = 3$. Find t when $d = 125$.

$k = 5$, $t^2 = 25$, $t = 5$

Platinum

$y \propto \sqrt{x}$. $y = 12$ when $x = 9$. Find y when $x = 49$.

$k = 4$, $y = 28$

Exam question 1

4 marks

P is directly proportional to the square of Q. P = 72 when Q = 6. Find a formula for P and work out P when Q = 4.

Space for working

Mark scheme: $P = 2Q^2$, $P = 32$

Objective 1 achieved

Solve direct proportion problems



Objective 2

Solve inverse proportion problems

✓ Achieved: Solve direct proportion problems

Key idea

Learn

!

$y \propto 1/x$ means $y = k/x$. As one goes up, the other goes down.

Objective 2: Solve inverse proportion problems

I do: example 1

I do

$y \propto 1/x$. $y = 6$ when $x = 4$. Find y when $x = 3$.

1. Write $y = k/x$
2. Substitute: $6 = k/4$, so $k = 24$
3. Formula: $y = 24/x$
4. When $x = 3$: $y = 24 \div 3 = 8$

I do: example 2

I do

$y \propto 1/x^2$. $y = 5$ when $x = 2$. Find y when $x = 4$.

1. Write $y = k/x^2$
2. Substitute: $5 = k/4$, so $k = 20$
3. Formula: $y = 20/x^2$
4. When $x = 4$: $y = 20 \div 16 = 1.25$

I do: example 3

I do

4 workers take 9 days. How long for 6 workers?

1. More workers → fewer days (inverse proportion)
2. Total work = 4 × 9 = 36 worker-days
3. Share between 6 workers: 36 ÷ 6
4. = 6 days

Mini whiteboards

We do

1. $y \propto 1/x$, $y = 10$ when $x = 3$. y when $x = 5$?

Answer: 6

2. $y \propto 1/x$, $y = 4$ when $x = 6$. y when $x = 8$?

Answer: 3

3. $y \propto 1/x^2$, $y = 9$ when $x = 2$. y when $x = 3$?

Answer: 4

4. 4 workers take 6 days. How long for 3 workers?

Answer: 8 days

You do: choose your challenge

You do

Bronze

$y \propto 1/x$. $y = 8$ when $x = 5$. Find y when $x = 10$.

Silver

$y \propto 1/x^2$. $y = 2$ when $x = 6$. Find y when $x = 3$.

Gold

5 pumps empty a tank in 12 hours. How long would 3 pumps take?

Platinum

$I \propto 1/d^2$. What happens to I when d is doubled?

You do: answers

Check

Bronze	4
Silver	8
Gold	20 hours
Platinum	I is divided by 4 (a 75% decrease)

If you finish early

Extra

Bronze

$y \propto 1/x$. $y = 6$ when $x = 4$. Find y when $x = 12$.

2

Silver

$y \propto 1/x$. $y = 5$ when $x = 8$. Find x when $y = 10$.

$k = 40$, $x = 4$

Gold

$y \propto 1/x^2$. $y = 10$ when $x = 3$. Find y when $x = 5$.

$k = 90$, $y = 3.6$

Platinum

$F \propto 1/d^2$. What happens to F when d is halved?

F is multiplied by 4

Exam question 1

3 marks

F is inversely proportional to d^2. F = 20 when d = 3. Find F when d = 6.

Space for working

Mark scheme: k = 180, F = 180 ÷ 36 = 5

Exam question 2

2 marks

Find d when F = 45.

Space for working

Mark scheme: $d^2 = 4$, $d = 2$

Objective 2 achieved

Solve inverse proportion problems



Real life

Apply

A journey takes 2.5 hours at 60 mph. How long does it take at 75 mph?

Distance = 150 miles, 150 ÷ 75 = 2 hours

Extension: stretch yourself

Stretch

1. $y \propto x^2$. x increases by 50%. By what percentage does y increase?

$1.5^2 = 2.25 \rightarrow$ 125% increase

2. $y \propto 1/x$. What happens to y when x is doubled?

y is halved

Exit ticket: exam questions

Exit

1. $y \propto x$. $y = 21$ when $x = 7$. Find y when $x = 12$. **(2 marks)**

2. $y \propto \frac{1}{x}$. $y = 9$ when $x = 4$. Find y when $x = 6$. **(3 marks)**

3. $T \propto \sqrt{L}$. $T = 6$ when $L = 9$. Find T when $L = 25$. **(3 marks)**

Exit ticket: mark scheme

Mark it

Q1 36 [2]

Q2 6 [3]

Q3 k = 2, T = 10 [3]

Total: ____ / 8

Today I achieved

- ✔ **Solve direct proportion problems**
- ✔ **Solve inverse proportion problems**

Rate yourself: 1 = still unsure 2 = getting there 3 = confident

Homework: exam questions

Due next lesson

1. $y \propto x$. $y = 28$ when $x = 4$. Find y when $x = 9$. [2]

2. $A \propto r^2$. $A = 45$ when $r = 3$. Find A when $r = 6$. [3]

3. $y \propto 1/x$. $y = 12$ when $x = 3$. Find y when $x = 9$. [3]

4. $F \propto 1/d^2$. $F = 50$ when $d = 2$. Find d when $F = 8$. [3]

5. 8 workers build a wall in 15 days. How long would 12 workers take? [2]

Extra practice: Maths Genie → Edexcel question bank → Direct and Inverse Proportion